

Frontend Product Requirements Document (PRD)

Project: Obsidian

Version: v1.0

Scope: Complete Frontend Architecture & UX System

Platform Targets:

- Web Application (Primary)
- Progressive Web App (PWA)
- Desktop Wrapper (Future Electron/Tauri)
- Mobile Responsive Web

Reference Backend Architecture:

1. Frontend Vision

Product Objective

Obsidian's frontend should provide a premium, ultra-smooth, Discord-inspired virtual office experience specifically optimized for:

- Indian startup founders
- Remote engineering teams
- Gaming-style collaborative workflows
- Persistent communication
- Lightweight low-bandwidth usability
- Fast onboarding

The frontend must feel:

- Fast
- Minimal

- Professional
- Immersive
- Real-time
- Interactive

while remaining highly optimized for:

- Low-end laptops
 - Mid-range Android devices
 - Unstable Indian internet conditions
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2. Frontend Core Principles

2.1 UX Philosophy

The UX should prioritize:

Principle	Meaning
Instant Interaction	No delayed UI reactions
Persistent Context	Users never lose navigation state
Low Cognitive Load	Minimal clutter
Real-Time Presence	Workspace always feels alive
Voice-First Collaboration	Voice rooms accessible in 1 click
Keyboard Efficiency	Fast productivity workflows

3. Frontend Technology Stack

Layer	Technology
Framework	Next.js 15
UI Library	React 19

Styling	Tailwind CSS
State Management	Zustand
Server State	TanStack Query
Animations	Framer Motion
Voice SDK	Amazon Chime SDK
Authentication	Firebase Auth
Database Realtime	Firestore SDK
Forms	React Hook Form + Zod
Icons	Lucide React
Notifications	Sonner
Routing	App Router
PWA	next-pwa
Charts	Recharts
File Uploads	AWS S3 Signed Upload
Theme Engine	next-themes

4. Frontend Application Architecture

4.1 App Structure

```
app/  
├── auth/  
├── onboarding/  
├── dashboard/  
├── workspace/  
├── settings/  
└── billing/
```

— channels/
— voice/
— admin/
— api/

4.2 Frontend Architecture Pattern

Hybrid Architecture

The frontend uses:

- SSR (SEO + performance)
 - CSR (real-time interactions)
 - Streaming UI
 - Suspense-based loading
 - Incremental rendering
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5. Authentication Experience

5.1 Login Flow

User Journey

Landing Page



Sign In



Firebase Authentication



Workspace Selection



Dashboard Redirect

5.2 Supported Login Methods

Method	Priority
Email/Password	Primary
Google OAuth	Primary
GitHub OAuth	Secondary
Magic Link	Future

5.3 Frontend Authentication States

State	UI Behavior
Loading	Skeleton UI
Authenticated	Workspace access
Unauthenticated	Redirect to login
Suspended	Billing warning
Expired Session	Silent token refresh

6. Landing Page PRD

6.1 Landing Page Goals

The landing page must:

- Communicate premium feel instantly
- Explain virtual-office concept clearly

- Drive free-trial conversion
 - Build trust with startups
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6.2 Landing Page Sections

Hero Section

Components

- Animated background
 - Workspace mockup
 - CTA buttons
 - Floating voice UI demo
 - Startup-focused headline
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Features Section

Cards

- Persistent Voice Rooms
 - Team Collaboration
 - Workspace Management
 - Role Permissions
 - Startup Scaling
-

Pricing Section

Interactive pricing cards:

Tier	CTA
Gold	Start Small
Premium	Most Popular

Footer

Includes:

- Social links
 - Terms
 - Privacy
 - API status
 - Support
-

7. Workspace Dashboard PRD

7.1 Dashboard Layout

Top Navigation	
Sidebar	Main Workspace
Channels	Messages
Members	Voice Rooms
Billing	Activity Feed

7.2 Sidebar Components

Sections

- Workspace Switcher
- Text Channels
- Voice Channels
- Members Online

- Direct Messages
 - Notifications
 - Billing
 - Settings
-

7.3 Main Content Areas

Dynamic rendering based on selected route:

Route	View
/chat	Messaging
/voice	Voice Rooms
/files	File Browser
/analytics	Workspace Insights
/settings	Settings Panel

8. Messaging System PRD

8.1 Messaging Features

Core Features

- Real-time messaging
- Markdown support
- File attachments
- Mentions
- Replies
- Reactions
- Pinned messages
- Infinite scroll

8.2 Message UI Structure

Avatar
Username
Timestamp
Message Content
Attachments
Reactions
Reply Thread

8.3 Message States

State	UI
Sending	Opacity animation
Sent	Normal
Failed	Retry indicator
Edited	Edited label
Deleted	Soft placeholder

9. Voice Channel Experience

9.1 Voice UX Goals

The voice experience should feel:

- Instant
- Frictionless
- Natural
- Persistent

similar to Discord.

9.2 Voice Channel Layout

Voice Header
Participants Grid
Controls Bar
Activity Indicators
Connection Stats

9.3 Voice Controls

Control	Function
Mute	Toggle microphone
Deafen	Disable incoming audio
Disconnect	Leave channel
Push-To-Talk	Optional
Noise Suppression	AI filtering

9.4 Voice States

State	UI Behavior
Connecting	Animated loader
Connected	Active waveform
Muted	Red mic icon
Speaking	Green outline
Poor Network	Warning badge

10. Presence & Real-Time Status

10.1 Presence Indicators

Status	Color
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Online	Green
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Away	Yellow
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Busy	Red
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Offline	Gray
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10.2 Real-Time Presence Features

- Live typing indicators
 - Active voice indicators
 - Online member counts
 - Real-time workspace updates
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11. Workspace Creation Flow

11.1 Onboarding Steps

Create Account



Create Workspace



Select Industry



Choose Tier

↓
Invite Team
↓
Workspace Ready

11.2 Workspace Setup UI

Forms

- Workspace Name
 - Team Size
 - Industry Type
 - Invite Emails
-

12. Billing & Subscription UX

12.1 Billing Dashboard

Components

- Current Plan
 - Storage Usage
 - Team Usage
 - Upgrade CTA
 - Invoice History
-

12.2 Razorpay Checkout Flow

Select Plan
↓
Checkout Modal
↓

Payment Verification



Success Animation



Workspace Upgraded

13. File Management PRD

13.1 File Browser Features

- Drag-and-drop uploads
 - Folder support
 - Search
 - Preview mode
 - Upload progress
 - File sharing
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13.2 Upload UX

States

State	UI
Uploading	Progress bar
Processing	Spinner
Complete	Success toast
Failed	Retry option

14. Admin & Role Management UI

14.1 Admin Dashboard

Features

- Role editor
 - Permissions matrix
 - Team analytics
 - Invite management
 - Workspace moderation
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14.2 Role Permission Matrix

	Admin	Developer	HR
Send Messages	✓	✓	✓
Join Voice	✓	✓	✓
Delete Msgs	✓	x	x
Manage Roles	✓	x	x

15. Notifications UX

15.1 Notification Types

Type	Priority
Mentions	High
Voice Invites	High
Billing	Medium
System Updates	Low

15.2 Notification Delivery

- In-app toast
 - Push notifications
 - Badge counters
 - Email alerts
-

16. Mobile Responsive PRD

16.1 Mobile Objectives

The mobile experience should support:

- Quick communication
 - Voice participation
 - Notifications
 - Team awareness
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16.2 Mobile Layout Adaptation

Desktop	Mobile
Sidebar	Drawer
Multi-panel	Single panel
Full grid	Stack layout

17. Performance Optimization

17.1 Frontend Performance Goals

Metric	Target
First Paint	< 1.5s
Time To Interactive	< 3s
Voice Join Delay	< 1s
Message Latency	< 300ms

17.2 Optimization Techniques

UI

- Lazy loading
- Virtualized lists
- Dynamic imports
- Memoization

Network

- Request batching
 - Firestore caching
 - CDN assets
 - Optimistic updates
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18. Animation System

18.1 Motion Philosophy

Animations must feel:

- Fast
- Smooth
- Minimal
- Responsive

18.2 Animation Usage

Action	Animation
Modal Open	Scale fade
Message Send	Slide up
Voice Join	Glow pulse
Notifications	Soft slide

19. Design System PRD

19.1 Visual Theme

Primary Style

- Dark premium interface
 - Cyber-office aesthetic
 - Startup-focused branding
 - Glassmorphism accents
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19.2 Color System

Purpose	Color
Background	#0B0F19
Surface	#111827
Accent	#6D5EF5

Success #10B981

Error #EF4444

19.3 Typography

Type	Font
Headings	Inter
Body	Inter
Code	JetBrains Mono

20. Accessibility Requirements

20.1 Accessibility Standards

Must comply with:

- WCAG AA
 - Keyboard navigation
 - Screen reader support
 - Focus visibility
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20.2 Accessibility Features

- High contrast mode
 - Reduced motion
 - Keyboard shortcuts
 - Voice accessibility
-

21. Error Handling UX

21.1 Error Categories

Type	UX Response
Network	Retry
Auth	Re-login
Voice Failure	Reconnect
Upload Failure	Resume

21.2 Offline Handling

Features:

- Offline indicators
 - Cached messages
 - Retry queue
 - Auto resync
-

22. Frontend Security Requirements

22.1 Client Security

- Secure token storage
 - XSS sanitization
 - CSP headers
 - File validation
 - Anti-spam throttling
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22.2 Sensitive Operations

Require re-authentication for:

- Billing changes
 - Workspace deletion
 - Owner transfer
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23. Analytics & Tracking

23.1 Product Analytics

Track:

- Daily active users
 - Voice usage
 - Channel activity
 - Conversion rates
 - Invite acceptance
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23.2 Heatmaps & UX Insights

Future integrations:

- PostHog
 - Mixpanel
 - Hotjar
-

24. Frontend Scalability Strategy

24.1 Rendering Scalability

Support:

- 1000+ channels
 - 10,000+ messages
 - Large member lists
 - High-frequency updates
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24.2 State Optimization

Use:

- Zustand slices
 - Query caching
 - Selective rerenders
 - WebSocket batching
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25. PWA Requirements

25.1 PWA Features

- Installable app
 - Push notifications
 - Offline support
 - Native-like feel
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25.2 Desktop-Like Experience

Features:

- Keyboard shortcuts

- Multi-panel layout
 - Persistent sessions
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26. Future Frontend Roadmap

Phase 2

- AI Meeting Assistant
- Video Calls
- Shared Screens

Phase 3

- Collaborative Whiteboards
- AI Workspace Insights
- Workspace Templates

Phase 4

- Plugin Marketplace
 - Third-Party Integrations
 - AI Moderation
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27. Frontend QA Requirements

27.1 Testing Layers

Layer	Tool
Unit Testing	Vitest

UI Testing Playwright

Component
Testing Storybook

E2E Testing Cypress

27.2 Critical Test Cases

- Voice reconnection
 - Real-time sync
 - Payment flows
 - Role permission enforcement
 - Upload failures
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28. Final Frontend Summary

The Obsidian frontend is designed to deliver:

- A premium startup collaboration experience
- Real-time immersive communication
- Discord-like voice collaboration
- Scalable workspace infrastructure
- Mobile-responsive productivity
- High-performance interaction systems

The frontend architecture emphasizes:

- Real-time synchronization
- Ultra-low latency interactions
- Minimal cognitive load
- Modern startup aesthetics
- Production-grade scalability
- Efficient rendering performance

This frontend PRD establishes the complete product blueprint required for:

- UI/UX design
- Frontend development

- Component architecture
- State management
- Real-time interaction systems
- Production deployment readiness